



Research article

Spatio-temporal observation of atmospheric CO₂ concentrations over Bangladesh: A remote sensing approach

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ABSTRACT

Climate change poses severe and potentially irreversible threats to developing countries, particularly Bangladesh. Atmospheric carbon dioxide (CO₂) plays a significant role in accelerating these adverse impacts. Effective spatio-temporal observation of atmospheric CO₂ concentrations using a remote sensing approach can help address these challenges. This study utilized dry-air mole fraction of CO₂ (XCO₂) observations from Orbiting Carbon Observatory-2 (OCO-2) to monitor CO₂ concentrations in Bangladesh. XCO₂ anomalies (MXCO₂) were calculated to detrend and deseasonalize the retrievals. The spatio-temporal distributions of CO₂ concentrations and relevant factors were analyzed within divisional boundaries. Pearson's correlation and Spearman's rank correlation were performed to identify the factors influencing the variation of CO₂ concentrations. The climate vulnerability ranking of divisions was determined based on atmospheric CO₂ concentrations and anthropogenic emissions. The spatio-temporal distribution of CO₂ concentrations showed a rising trend with seasonal fluctuations. Nationwide, CO₂ concentrations increased by 3.67 % from 2015 to 2021. The anthropogenic CO₂ emissions were identified as a significant factor contributing to the increase in CO₂ concentrations. The mean tropospheric NO₂, population density, and land-use change also exhibited a strong correlation with changes in CO₂ concentrations. Meteorological factors such as temperature, rainfall, and wind speed did not demonstrate any discernible trend or correlation. Based on atmospheric CO₂ concentration and emission rates, Dhaka was identified as the most vulnerable division, which serves as a central industrial hub, while Barishal was identified as the least vulnerable division due to its lower level of industrial activity. This study demonstrates the effectiveness of the remote sensing approach for monitoring CO₂ and observing relevant factors to explain climate vulnerability in Bangladesh. The findings of this study can contribute to informed decision-making regarding effective environmental measures and sustainable urban planning.

1. Introduction

Climate change impacts are imposing great challenges worldwide with increasing intensity over time, accelerated by anthropogenic greenhouse gas emissions [1]. Climate change causes risks such as sea level rise, extreme weather events, and impacts on biodiversity. These challenges make South Asian countries, especially Bangladesh, vulnerable. This vulnerability is exacerbated by

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their geographic location, socio-economic condition, limited resources, weak infrastructure, high population density, and insufficient governance [2]. The major contributors to such risks are atmospheric carbon dioxide (CO_2) emissions and higher concentrations due to increased fossil fuel burning from growing industrialization and urbanization [3]. The global mean atmospheric CO_2 concentration is 425 ppm, which has been increasing ever since the Industrial Revolution [4]. This concentration has increased by approximately 50 % in the last 200 years due to anthropogenic activities [5]. A similar rate of increase may cause global warming of 1.5°C by the middle of this century [6]. Bangladesh is highly vulnerable to climate-induced risks [7] and was ranked seventh in the Climate Risk Index 2021 [8]. Moreover, it has the fifth-worst air quality in the world [9]. Therefore, the spatio-temporal observation of atmospheric CO_2 concentrations and relevant factors is important for a comprehensive understanding of the trend.

Atmospheric CO_2 is an important greenhouse gas that is colorless, odorless, and non-flammable. It absorbs and radiates heat and contributes significantly to climate change and global warming [5,10]. The long atmospheric lifetime of CO_2 (years) allows atmospheric dynamics to exert a profound influence on its spatio-temporal distribution. Elevated concentrations of atmospheric CO_2 can cause a ripple effect across the planet, affecting various aspects, including climate and ecosystems [5,11–13]. The inclining trend of atmospheric CO_2 worldwide necessitates accurate measurement of atmospheric concentrations. Over the last two decades, significant advancements in spaceborne measurements of air pollutants have led to more accurate atmospheric composition monitoring, providing global-scale pollution information [14–17]. However, CO_2 has a relatively high background concentration and a strong seasonal cycle, intensifying anthropogenic CO_2 signature identification by remote sensing applications.

Remote sensing applications are essential for accurate regional and global scale monitoring of atmospheric CO_2 , as ground station-based measurements are hindered by limited spatial coverage and uneven distribution [18–20]. Satellite-based observations have improved atmospheric CO_2 monitoring by covering large regions and providing ample sampling data, which can be attained using

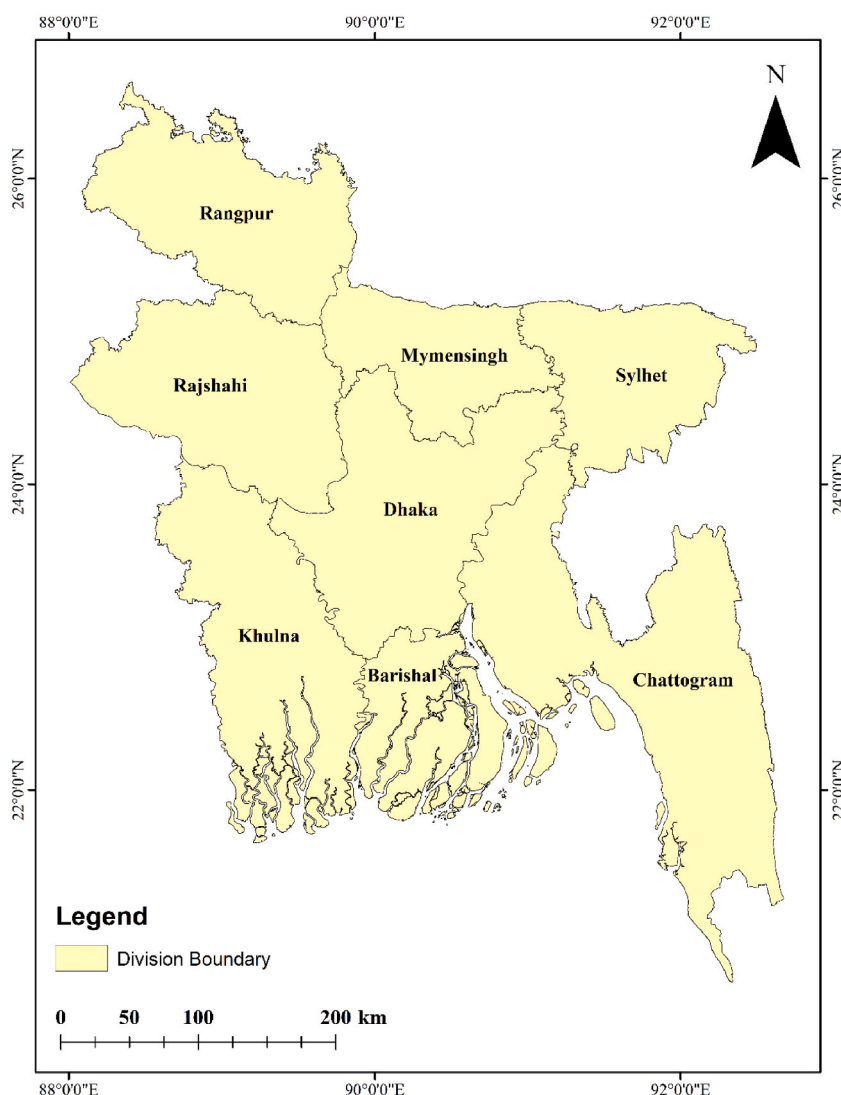


Fig. 1. Map of the study area showing the eight divisions of Bangladesh.

various satellite sensors [21–23]. The thermal infrared (TIR) bands used in these sensors have reduced sensitivity in the lower troposphere where CO₂ emission and sink take place mostly. The Scanning Imaging Absorption Spectrometer for Atmospheric Cartography (SCIAMACHY) on ENVISAT was the first instrument to detect atmospheric CO₂ signals. It provided reliable global-scale observations of the column-averaged dry-air mole fraction of CO₂ (XCO₂) [24,25]. The Greenhouse Gases Observing Satellite (GOSAT), launched by Japan in 2009, was the first satellite for atmospheric CO₂ and CH₄ monitoring [26]. GOSAT has been used to detect CO₂ emission signatures from megacities [27] and to compare atmospheric CO₂ abundance with emission inventories [28]. The necessary observation of CO₂ emission and concentration can be improved with better spatio-temporal resolution and coverage, as concluded from recent studies.

High-resolution observations of atmospheric CO₂ are available as XCO₂ readings, obtained from the Orbiting Carbon Observatory 2 (OCO-2) satellite, launched by the National Aeronautics and Space Administration (NASA) in 2014 [29]. OCO-2 is the first dedicated satellite for monitoring and identification of CO₂ pollution, and the derived data are used for regional and local scale studies due to precise measurements [30,31]. The utilization of the OCO-2 dataset is still very recent for studies worldwide. Previous studies in Bangladesh have investigated the relationship between CO₂ emission and energy consumption [32–34], identified determinants of CO₂ emission [35], analyzed emissions from pollution sources [36,37], and estimated the pattern and distribution of particulate matter [38] and trace gases [39]. These studies employed data from emission inventories to study anthropogenic CO₂ emission by satellite observations. Although Bangladesh is vulnerable to climate change, there has been no remote sensing-based study to accurately monitor the atmospheric CO₂ concentrations until now. To fill the existing literature gap, this study provides a comprehensive spatio-temporal assessment of the CO₂ concentrations by employing the OCO-2 dataset. Furthermore, the factors influencing the pattern of CO₂ concentrations were identified and used for spatio-temporal comparison and correlation analysis for a broader understanding. Finally, the divisions of Bangladesh were ranked based on CO₂ concentrations and emissions to determine climate vulnerability. This study makes a valuable contribution to climate change research by demonstrating the trend of atmospheric CO₂ using OCO-2 data. It is one of the first investigations to study various factors influencing atmospheric CO₂ in Bangladesh. The findings of this study can aid urban planners and policymakers in making sustainable decisions, evaluating the effectiveness of existing policies, and more effectively addressing the issue.

2. Methodology

2.1. Study area

Bangladesh is a developing country in South Asia, sharing borders with India to the east, north, and west and Myanmar on the southeast. It is also the eighth most populous country in the world, with more than 170 million people. The land area of Bangladesh is 148,460 km². Bangladesh is one of the world's largest river deltas, formed by the confluence of the Ganges, Brahmaputra, and Meghna rivers. It is a riverine country containing 230 large rivers and their tributaries [40].

High temperatures and heavy rainfall with fairly marked seasonal variations characterize the climate of Bangladesh. The mean temperature varies between 24 and 30 °C, while the mean annual rainfall is 2500 mm [41]. Bangladesh is divided into eight major administrative units called divisions, which are considered for this study, as shown in Fig. 1. The eight divisions are Dhaka, Chattogram, Khulna, Barishal, Rangpur, Rajshahi, Sylhet, and Mymensingh. Among these, Chattogram is the largest, and Mymensingh is the smallest. Dhaka is the most populous division with the highest industrial activities, while Barishal has the lowest industrial activities [42].

2.2. Identification of factors influencing atmospheric CO₂ concentrations

From the literature review, eight factors were identified as relevant to atmospheric CO₂ concentration, including anthropogenic CO₂ emissions [3], tropospheric NO₂ [43], population density [44], vegetation [45], built-up area [46], temperature [47], rainfall [48], and wind speed [49]. As indicated in Table S1, these factors play a role in both increasing and decreasing CO₂ concentrations in the atmosphere.

2.3. Data collection

Data of the eight factors relevant to atmospheric CO₂ concentration were collected to investigate their spatio-temporal distribution and conduct correlation analysis during the study period from 2015 to 2021. The data collection and extraction methods are described below.

2.3.1. Atmospheric CO₂ concentrations

The atmospheric CO₂ concentrations were retrieved using XCO₂ data from the OCO-2 satellite. It is in a sun-synchronous polar orbit with a sampling time of 13.36 LT [50]. This satellite has a 16-day repeat cycle on a set of 233 orbit paths. The spatial resolution of OCO-2 is about 1.3 × 2.25 km at the nadir. It has been in service since September 6, 2014, and has provided around 1 million observations daily since then [51]. The XCO₂ data were retrieved using the Atmospheric CO₂ Observations from Space (ACOS) retrieval algorithm [50]. These data have been validated as consistent and reliable for atmospheric CO₂ monitoring due to sounding precision against accurate measurements [52]. The latest ACOS/XCO₂ Lite product version (v10r) [53] from the EARTHDATA website (https://cmr.earthdata.nasa.gov/virtual-directory/collections/C1685783927-GES_DISC, accessed on 7 February 2024) was used in this study.

The data of the lowest quality were screened using quality flags set to zero.

Atmospheric CO₂ concentration measurement using a remote sensing approach is difficult due to its long atmospheric life and relatively large background concentrations. Therefore, the retrieved XCO₂ varies only about 2 % from pole to pole over the seasonal cycle [3]. This necessitates deseasonalizing and detrending the retrieved data to determine the concentration of atmospheric CO₂ by calculating anomalies of XCO₂ as shown in Equation (1). This process also helps to minimize the effect of the changing spatial distribution of the data points and regional-scale bias impacts [54].

$$MXCO_2 = XCO_{2(ind)} - XCO_{2(dm)} \quad (1)$$

Where, $MXCO_2$ indicates XCO₂ anomaly, $XCO_{2(ind)}$ denotes the individual XCO₂ retrievals in a day, and $XCO_{2(dm)}$ represents the XCO₂ daily median value. The daily median was calculated when at least 100 measurements were observed for the divisions on the same day. Considering the daily median value helps to remove the seasonal variability and increasing trend of CO₂ concentrations. Positive values indicate CO₂ emission into the atmosphere, and negative values indicate CO₂ absorption on the surface. Furthermore, the monthly and the annual mean of XCO₂ and MXCO₂ were calculated within the boundaries of the eight divisions.

2.3.2. Anthropogenic CO₂ emissions

The anthropogenic CO₂ emissions data were obtained from the Open-source Data Inventory for Anthropogenic CO₂ (ODIAC) dataset. This is a high-resolution global data product for fossil fuel CO₂ emissions, developed in 2009 by the National Institute for Environmental Studies (NIES), Japan. ODIAC estimates and distributes CO₂ emissions by incorporating individual power plant profiles and nighttime light data [55]. The dataset is widely used by researchers worldwide in studies of various scales to investigate anthropogenic CO₂ emissions [3,56]. The 1 km × 1 km version of the ODIAC2022 (Latest) dataset [57] was used in this study, obtained from the Center for Global Environmental Research website (https://db.cger.nies.go.jp/dataset/ODIAC/DL_odiad2022.html, accessed on 7 February 2024). The emission data for administrative zones were extracted by performing the Zonal Statistics as Table tool using ArcGIS 10.8.1.

2.3.3. Tropospheric NO₂

The tropospheric NO₂ column data, measured by the Ozone Monitoring Instrument (OMI), was used in this study. OMI has operated from the Aura satellite of NASA since October 2004. The solar backscattered light is measured in the UV–visible light spectral range by OMI with a 13 × 24 km² spatial resolution at nadir with almost daily global coverage [58]. OMI/Aura Nitrogen Dioxide (NO₂) Total and Tropospheric Column 1-orbit L2 Swath 13x24 km V003 (OMNO2) [59] dataset was used in this study from the EARTHDATA website (<https://data.nasa.gov/dataset/omi-aura-nitrogen-dioxide-no2-total-and-tropospheric-column-1-orbit-l2-swath-13x24-km-v003-a0854>, accessed on 7 February 2024). The NO₂ column data for administrative zones were extracted by performing Zonal Statistics as Table tool using ArcGIS 10.8.1.

2.3.4. Population density

Population density data were incorporated in this study from LandScan population density data. Oak Ridge National Laboratory (ORNL) initiated the LandScan Program in 1997 to provide improved population estimation for consequent assessment, and since 1998, it has been updated annually [60]. The LandScan Global data were used in this study from the Oak Ridge National Laboratory website (<https://landscan.ornl.gov/>, accessed on 7 February 2024). The population density data for administrative zones were extracted by performing the Zonal Statistics as Table tool using ArcGIS 10.8.1.

2.3.5. Vegetation and built-up area

This study used remote sensing indices to determine the changing trend of land use. The two major land uses, vegetation and built-up area, were focused on as factors and identified by performing the Normalized Difference Vegetation Index (NDVI) and the Normalized Difference Built Index (NDBI) accordingly. Greenness and built surface change over the years were assessed using remote sensing indices. The Landsat 8 Level 2 Collection 2 data were used to determine the NDVI and the NDBI for the eight divisions from 2015 to 2021 in Google Earth Engine.

Amongst the several vegetation indices, NDVI is the most widely used index [61]. This index was chosen for this study as it can significantly improve green vegetation detection by using simple Vis combined with NIR and visible bands. NDVI calculation is performed according to Equation (2):

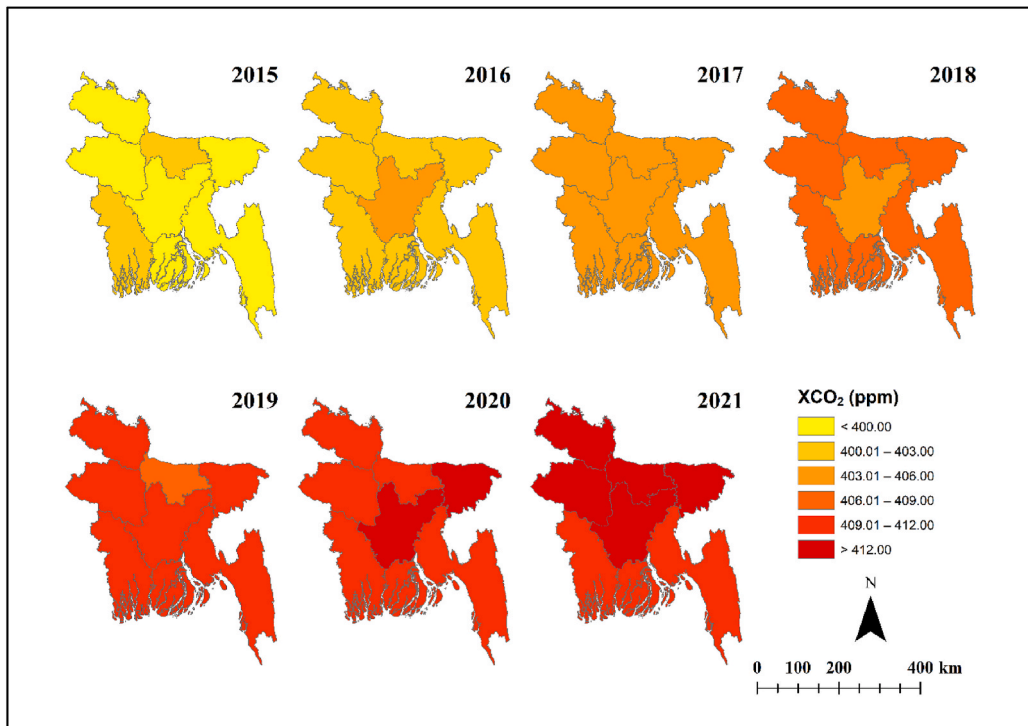
$$NDVI = \frac{NIR - Red}{NIR + Red} \quad (2)$$

Where, *NIR* indicates the *Near-infrared* band, and *Red* indicates the *Red* band. *NDVI* value varies from −1 to +1, with forestry and water being represented by values closer to +1 and values closer to −1 accordingly.

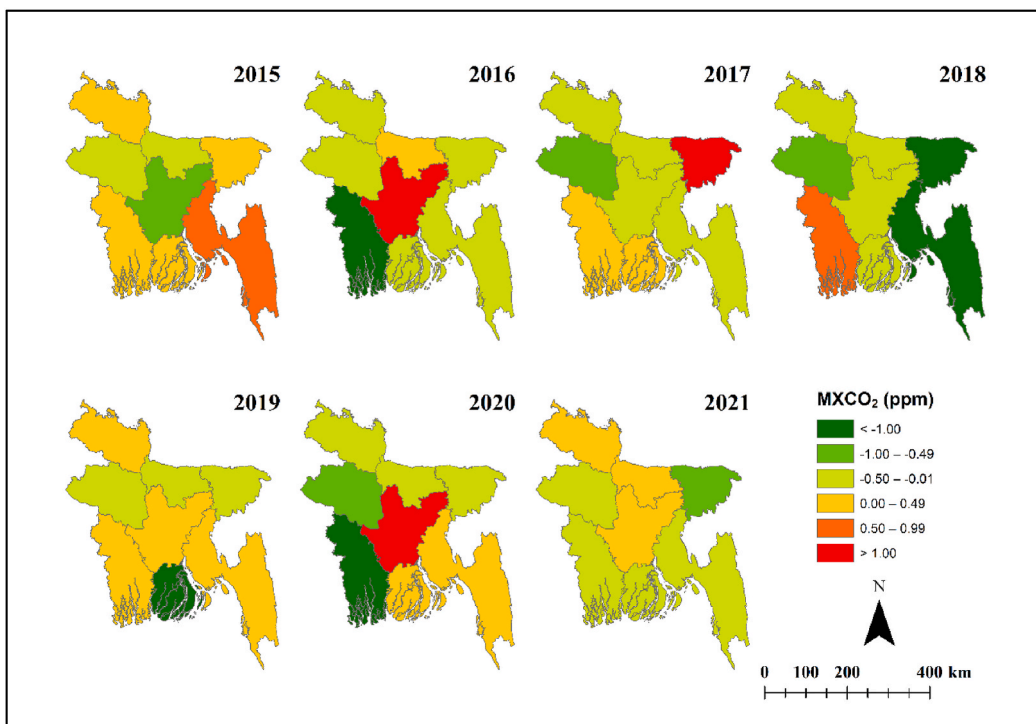
NDBI, widely used for mapping built-up or urban areas, obtains a 92.6 % precision level [62], which was adopted in this study. NDBI calculation is performed using Equation (3):

$$NDBI = \frac{NIR - SWIR}{NIR + SWIR} \quad (3)$$

Where, *NIR* indicates Near-infrared band and *SWIR* indicates Short-wave infrared band. The values of *NDBI* vary from −1 to +1. A



(a)



(b)

Fig. 2. Spatio-temporal distribution of (a) XCO_2 and (b) $MXCO_2$.

higher NDBI value indicates a built-up area, and a lower value indicates a non-built-up area.

2.3.6. Meteorological variables

The meteorological variables considered for this study were temperature, rainfall, and wind speed. The datasets for 2015–2021 were collected from the Bangladesh Meteorological Department (BMD), which provides historical meteorological data for its 34 established stations. Before data analysis, homogenization and data quality were assessed. Potential outliers were identified and filtered for correction. The daily data from NASA Power Access (<https://power.larc.nasa.gov/data-access-viewer/>, accessed on 7 February 2024) were collected for the three variables and compared with BMD data for validation purposes. Table Join was performed after collecting data with the weather station points. Further, Kriging Interpolation and Zonal Statistics as Table tools were used in ArcGIS 10.8.1 to extract data for all the administrative regions.

2.4. Data analysis and visualization

2.4.1. Spatio-temporal distribution

Mean XCO_2 and $MXCO_2$ values for Bangladesh were calculated on a monthly and seasonal basis to identify underlying trends. Three seasons were defined based on four months: Summer (March to June), Monsoon (July to October), and Winter (November to February). Furthermore, annual mean values of the two datasets and eight relevant factors were computed at the divisional level to examine their spatio-temporal distribution from 2015 to 2021.

2.4.2. Correlation analysis

Correlation analysis is a statistical method applied to determine the linear relationship between two categorical or quantitative variables and calculate their association by strength and direction. In this study, both the Pearson Correlation [63] and Spearman's Correlation [64] analyses were performed between $MXCO_2$ and the other identified relevant factors to clearly understand the current condition and trend of atmospheric CO_2 concentrations.

2.4.3. Climate-vulnerable regions ranking

The climate-vulnerable regions were identified considering the increment rate of atmospheric CO_2 concentration and anthropogenic CO_2 emission. Firstly, the rate of increment was calculated between 2015 and 2021 for the two factors. Then, the values were normalized for rank calculation. The divisions were ranked from 1 (most vulnerable) to 8 (least vulnerable).

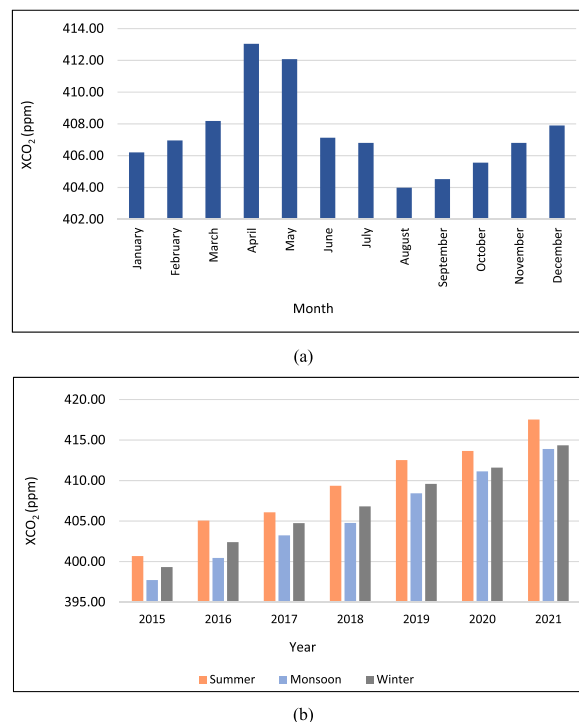


Fig. 3. Distribution of XCO_2 : (a) mean monthly distribution; (b) mean seasonal distribution.

3. Results and analysis

3.1. Spatio-temporal distribution of atmospheric CO₂ concentrations

The spatio-temporal distribution of CO₂ concentrations helps identify the vulnerable regions with higher concentrations and the trend of change in CO₂ concentrations. This study considered the XCO₂ readings to visualize the distribution. Fig. 2(a) presents the distribution of annual mean atmospheric CO₂ concentrations across different divisions of Bangladesh for the study period 2015–2021. The results indicate a gradual increase in atmospheric CO₂ concentration. In 2021, Dhaka, Rajshahi, Rangpur, Mymensingh, and Sylhet exhibited higher concentrations, exceeding 412 ppm, representing an increase of approximately 3.92 % since 2015. The highest mean concentration was observed for Dhaka at 416 ppm, and the lowest was 412 ppm for Barishal. Barishal, Chattogram, and Khulna showed CO₂ concentrations between 409 and 412 ppm in 2021, which increased by about 3.25 % from 2015. Overall, XCO₂ concentrations increased by 3.67 % across the country between 2015 and 2021.

The mean concentration range of all the divisions was the same during 2017 at 403–406 ppm. While Dhaka remained at the same range, the concentration increased for the rest of the divisions in 2018. The mean increase for the whole country was 3.67 % during this period. Table S2 presents the mean XCO₂ readings for the eight divisions during the study period. The net change in XCO₂ from 2015 to 2021 is shown in Fig. S4. The MXCO₂ distribution identifies the emission and absorption scenario of CO₂. The mean MXCO₂ over the years for different divisions is presented in Fig. 2(b). No discernible trends were observed for any of the divisions. The lowest values were observed for Rajshahi, with higher absorption over the years. The highest values were observed for Dhaka, indicating higher atmospheric CO₂ concentration and emission. The concentration of Rajshahi and Rangpur changed at a higher rate from 2020 to 2021. The absorption of Khulna reduced drastically from 2020 to 2021. A positive value indicates no atmospheric CO₂ absorption in Mymensingh in 2021, which has been negative since 2017.

Fig. 3(a) shows the monthly mean of XCO₂ for the whole country. The concentration shows an increasing trend from January to April, when it reaches the maximum concentration and starts declining. In August, the concentration is observed to be the lowest over the years. After that, the concentration starts increasing again, following the trend till the year ends. Fig. 3(b) shows that maximum concentrations over the years were observed during the summer season in April and May. The concentration tends to show a declining trend during the monsoon season. After the monsoon season, the concentration starts increasing again. The concentration in each season increased gradually every year compared to the previous years.

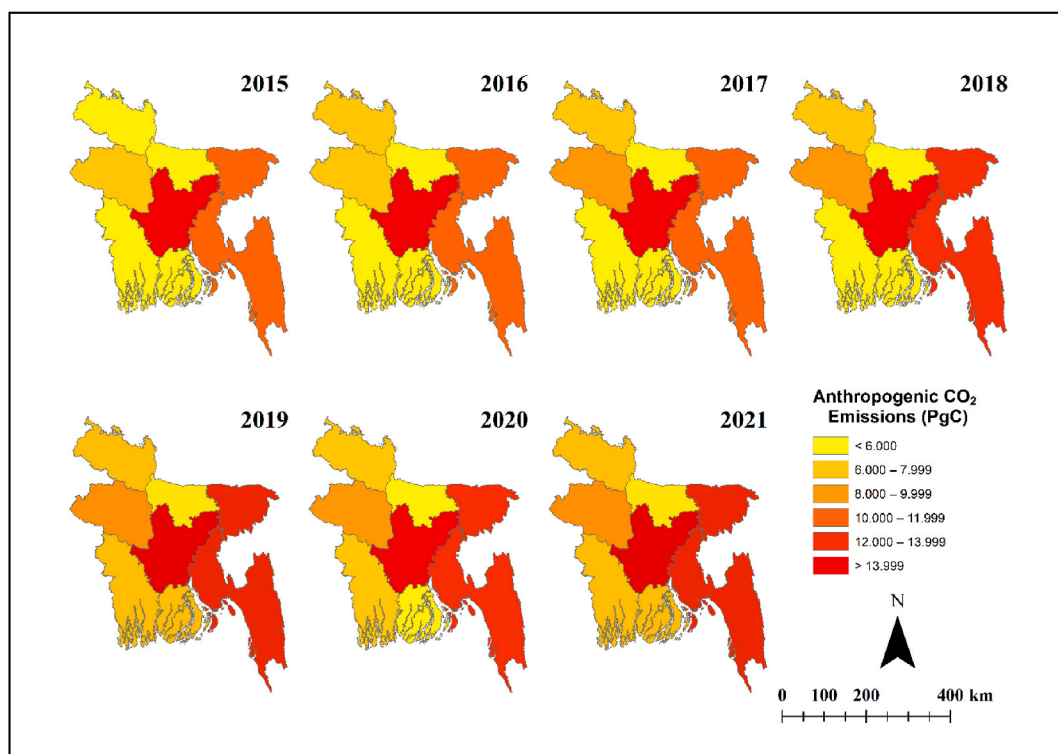


Fig. 4. Spatio-temporal distribution of anthropogenic CO₂ emissions.

3.2. Spatio-temporal distribution of anthropogenic CO₂ emissions

The anthropogenic CO₂ emissions in Fig. (4) show the emission pattern of regions over the study period. The emissions of Barishal remained below 6 PgC before 2019, increasing to 6.12 PgC in 2021. The emissions of Dhaka were significantly higher compared to the rest of the regions, increasing from 28.629 PgC in 2015 to 36.807 PgC in 2021. The emissions of Sylhet and Chattogram increased by 3.03 PgC and 3.05 PgC, respectively, from 2015 to 2021. The emissions of Mymensingh were 4.627 PgC in 2015, which remained below 6 PgC over the years. The emissions of Rajshahi increased by about 28.78 % during this period, while for Rangpur, the rate of change was 28.76 %. The highest and lowest rates of change of anthropogenic CO₂ emissions were observed for Dhaka (28.96 %) and Barishal (28.20 %), respectively. The spatio-temporal distribution of emissions complements the atmospheric CO₂ concentration pattern. The mean anthropogenic CO₂ emission readings for the eight divisions during the study period are mentioned in Table S3.

3.3. Spatio-temporal distribution of mean tropospheric NO₂

The mean tropospheric NO₂ distribution is important in tracing the anthropogenic emission scenario. The spatio-temporal NO₂ distribution in Fig. (5) depicts that the highest values were observed in Dhaka over the years, exceeding 280×10^{15} molec/cm². In contrast, the lowest values were observed in Chattogram, remaining below 160×10^{15} molec/cm² throughout the study period. NO₂ distribution decreased across all the divisions during 2019 and remained at a lower level during 2020. In 2021, the NO₂ distribution increased again. However, the overall national reading during this period was relatively lower than in previous years. Sylhet showed similarly low values to Chattogram, while Rajshahi exhibited a higher value throughout the period. The values of the divisions did not increase by more than 30×10^{15} molec/cm². The regions with higher mean tropospheric NO₂ distribution over the years, such as Dhaka and Rajshahi, exhibit higher CO₂ concentrations and emissions.

3.4. Spatio-temporal distribution of population density

The population density map in Fig. (6) shows that Dhaka has a significantly higher population density than the rest of the divisions. It had 2016 people/km² in 2021, which increased from 1899 people/km² in 2015. This density is much higher than the rest of the regions and almost thrice the density of the lowest-density region, Barishal. The population density of Barishal was 672 people/km² in 2015, which increased to 711 people/km² by 2021. It consistently remained the region with the lowest population density and also exhibited lower emissions over the years. The population densities of both Rajshahi and Mymensingh also increased to over 1150 people/km² in 2021. All the regions with higher population density were also associated with higher atmospheric CO₂ concentrations,

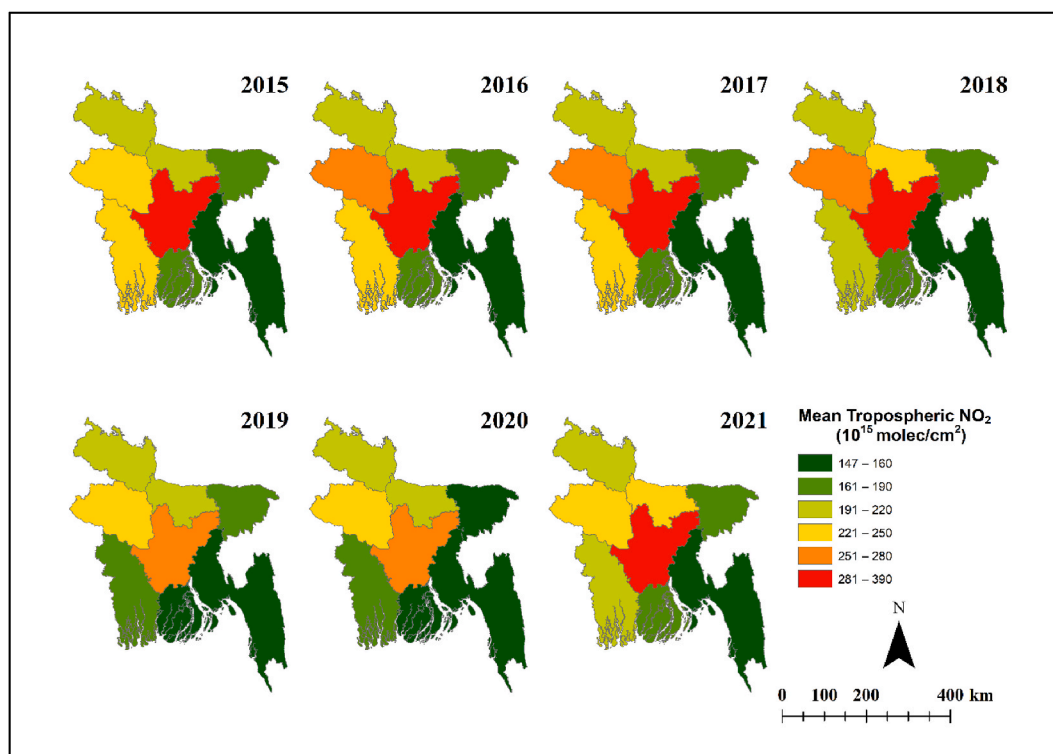


Fig. 5. Spatio-temporal distribution of mean tropospheric NO₂.

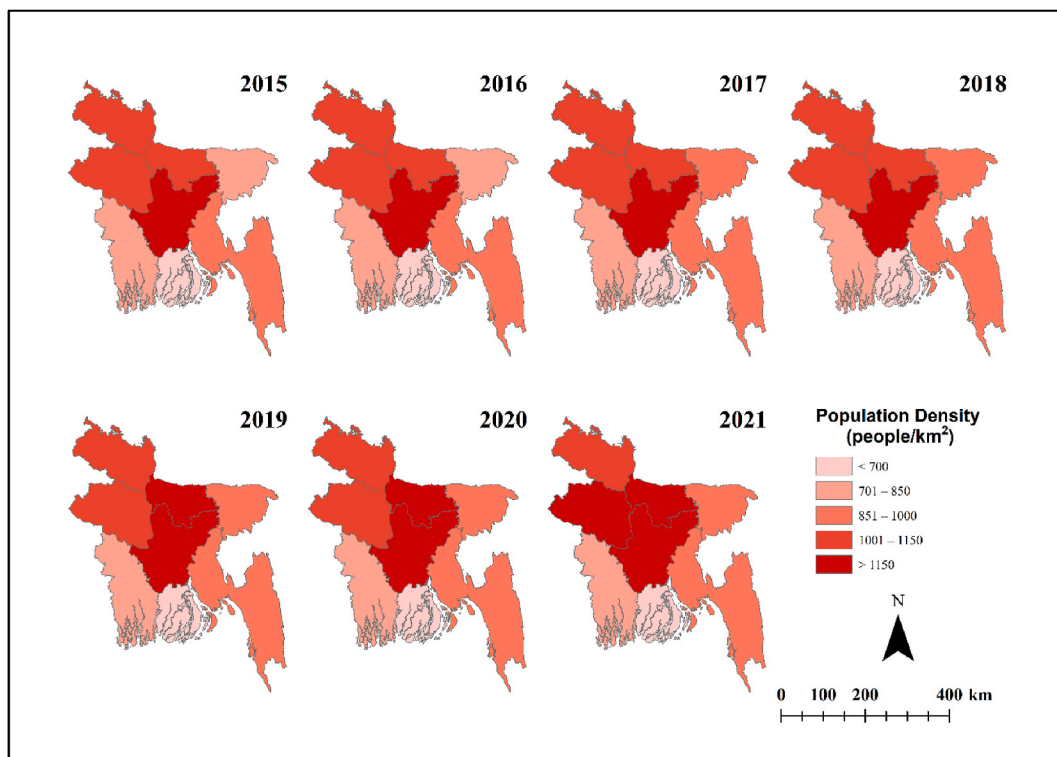


Fig. 6. Spatio-temporal distribution of population density.

as observed in Fig. (2). The mean increase in density was about 7 % for the whole country, where Dhaka had the highest rate of 12.26 %.

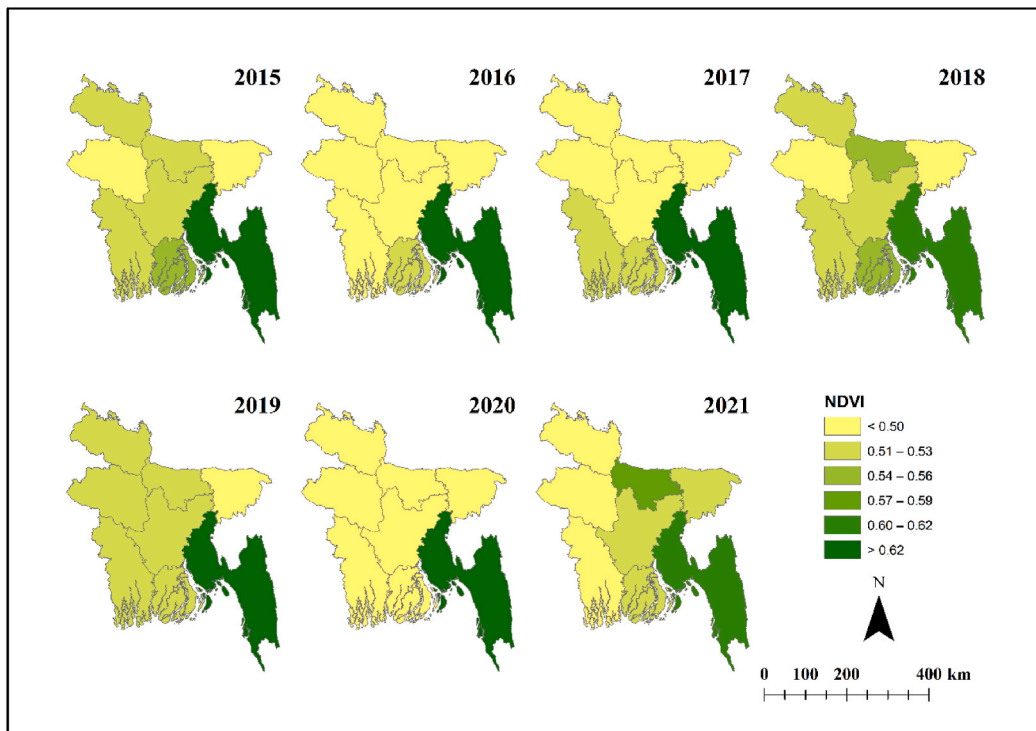
3.5. Spatio-temporal distribution of vegetation and built-up areas

The NDVI trend indicates the change in vegetation cover, which is important to detect regions with better CO₂ absorption. Fig. 7(a) shows the NDVI trend values for the eight divisions of Bangladesh between 2015 and 2021. Chattogram had the highest vegetation cover throughout the study period, with NDVI values exceeding 0.62. The value for Khulna and Barishal fluctuated in the 0.51–0.56 range during the period. During 2020, the values for all the divisions except Chattogram were below 0.50. NDVI values for Dhaka mostly ranged between 0.51 and 0.53, with limited deviations beyond this range. A slight decrease was observed in 2016–2017 and 2020, with values falling below 0.50 during these periods. Notably, some of these low NDVI regions coincided with areas of elevated atmospheric CO₂ concentrations, including Dhaka, Rajshahi, and Rangpur. The trend change slope of NDVI for Dhaka, Mymensingh, Rajshahi, and Rangpur displayed a declining trend for the study period, while the rest of the regions exhibited an inclining trend. Among the declining trends, Rangpur experienced the steepest decline, while Dhaka showed the smallest decline. Conversely, within the increasing trends, Chattogram displayed the largest increase and Khulna the smallest.

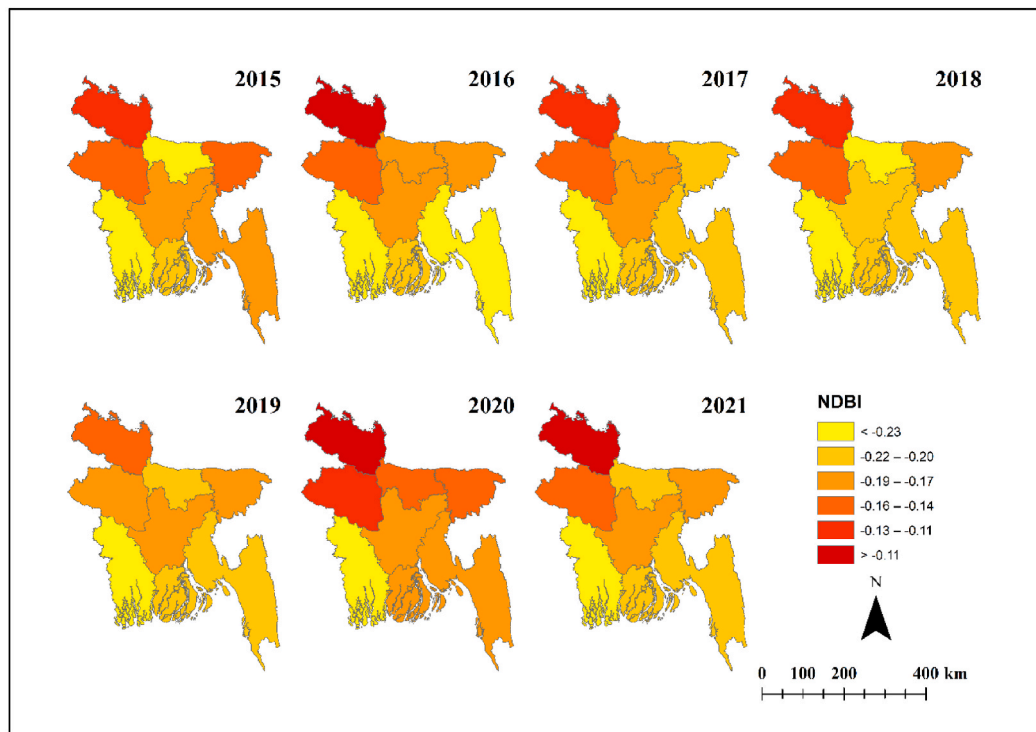
The NDBI trend helps to visualize the urbanization pattern, which is related to anthropogenic emissions, as shown in Fig. 7(b). Rangpur and Rajshahi were the regions with higher built-up area conversion, indicating rapid urbanization. Those regions also displayed higher atmospheric CO₂ concentrations. Throughout the study period, the Rangpur division consistently showed NDBI values above −0.11, with the lowest observed values ranging from −0.16 to −0.14 in 2020. The NDBI values for Rajshahi were mostly between −0.16 and −0.14. Conversely, Khulna demonstrated a consistently lower NDBI trend, remaining below −0.23 throughout the period. Dhaka exhibited values between −0.19 and −0.17, which were lower than those observed in Rajshahi and Rangpur. The trend change slope of NDBI for Rangpur showed the highest value of −0.12, while the lowest value of −0.26 was observed for Khulna in 2021.

3.6. Spatio-temporal distribution of meteorological variables

The annual mean temperature distribution between 2015 and 2021 of the divisions of Bangladesh is displayed in Fig. 8(a). While temperature fluctuations across most regions remained within approximately 0.5 °C, Mymensingh, Rangpur, and Sylhet consistently exhibited lower temperatures, remaining below 25.50 °C and cooler than other divisions throughout the study period. The temperature of Sylhet was mostly below 25 °C. In contrast, Barisal and Khulna experienced higher annual mean temperatures. Divisions with higher

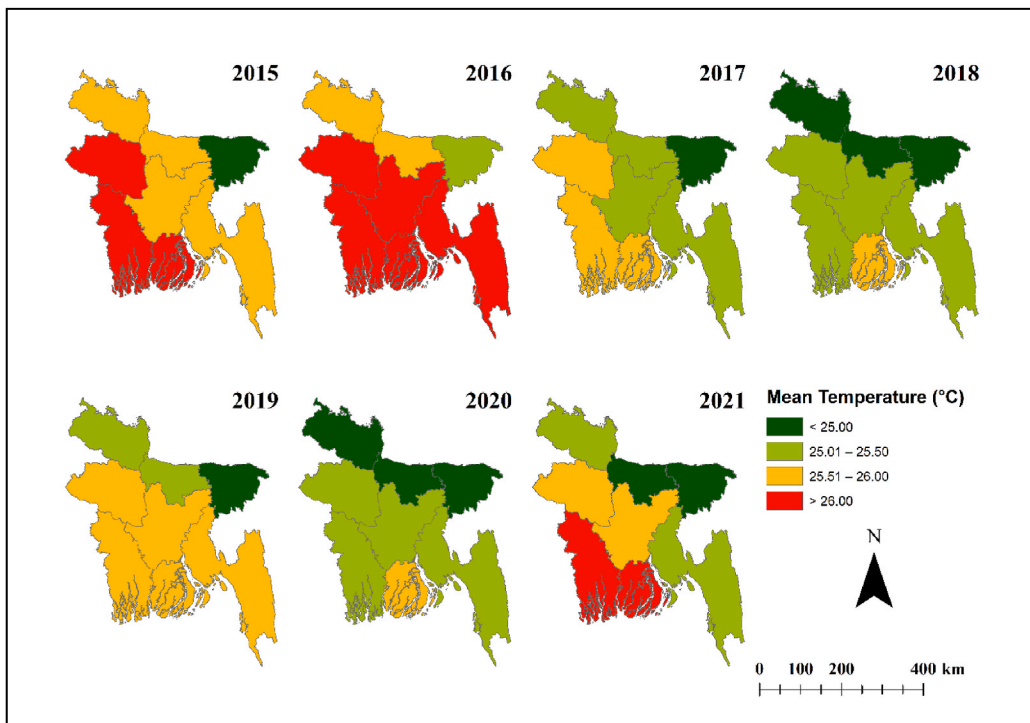


(a)

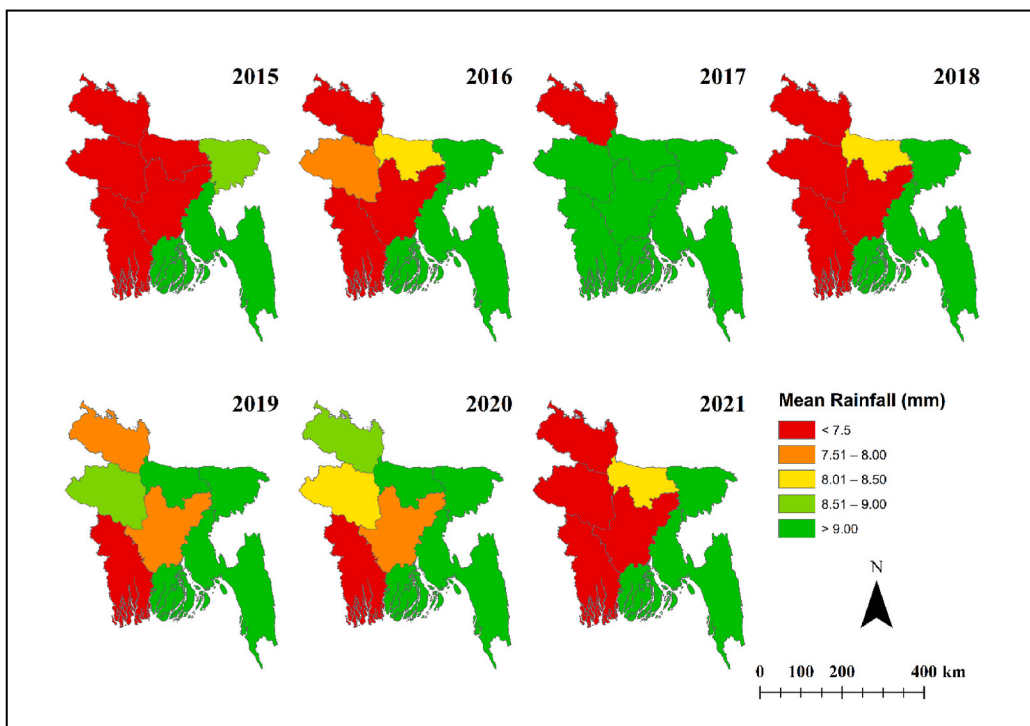


(b)

Fig. 7. Spatio-temporal distribution of land-use trends: (a) NDVI; (b) NDBI.

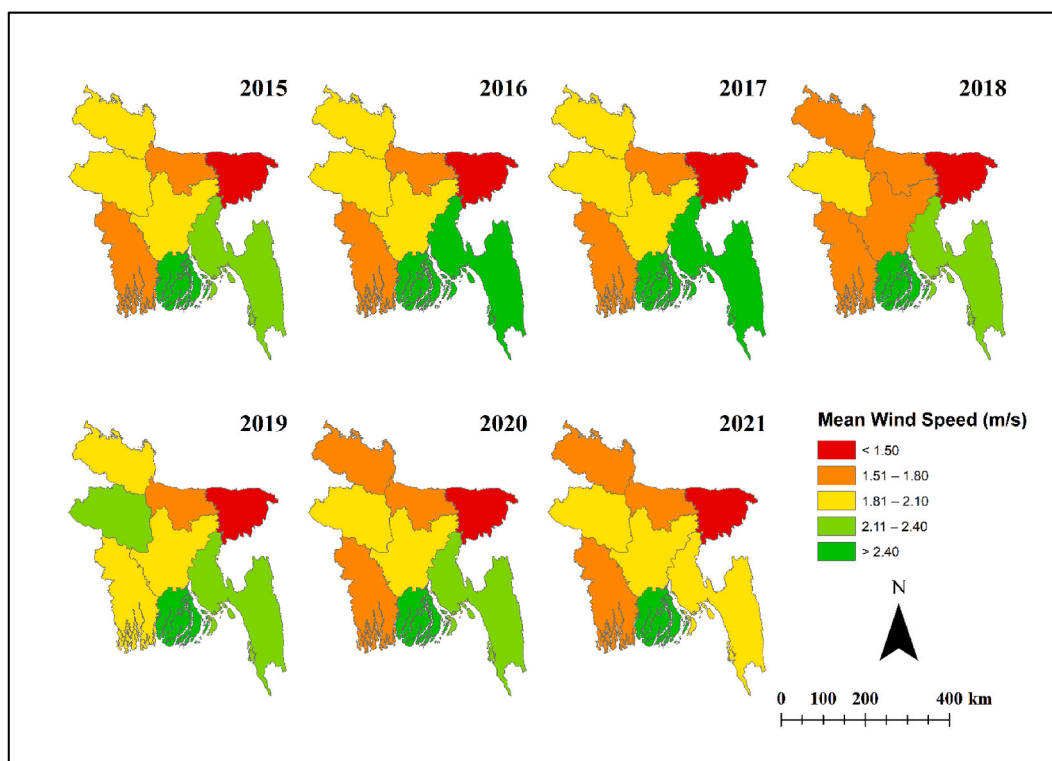


(a)



(b)

Fig. 8. Spatio-temporal distribution of meteorological variables: (a) mean temperature; (b) mean rainfall; (c) mean wind speed.



(c)

Fig. 8. (continued).

atmospheric CO₂ concentrations, such as Dhaka and Rajshahi, generally showed temperatures between 25 and 26 °C, particularly after 2017. Although the annual mean temperature did not display any distinct trend during this period, it remained relatively high until 2016, followed by a decrease from 2017 to 2018. In 2019, the temperature started to rise again. The annual mean temperature again dropped during 2020 when the COVID-19 pandemic occurred and started rising again in 2021.

Annual mean rainfall varied considerably across Bangladesh. Dhaka, Rajshahi, and Rangpur experienced lower annual rainfall, below 7.5 mm. Except for Rangpur, the rainfall for all the divisions reached a range above 9.00 mm per year. Mymensingh had a mean yearly rainfall ranging between 8.00 and 9.00 mm from 2016 to 2021. Barishal, Chattogram, and Sylhet had a mean annual rainfall above 9 mm during the period. As illustrated in Fig. 8(b), no clear increasing or decreasing trend in annual rainfall was observed. A notable increase in rainfall occurred between 2018 and 2019, followed by a period of relative stability until 2020 and then another increase in 2021. The regions with lower rainfall, such as Dhaka, Rangpur, and Rajshahi, exhibited higher atmospheric CO₂ concentrations compared to the other regions.

Fig. 8(c) shows the annual mean wind speed for the eight divisions from 2015 to 2021. Barisal consistently exhibited the highest annual mean wind speed, exceeding 2.40 m/s throughout the period. Chattogram showed fluctuating wind speeds, ranging from 2.11 to 2.40 m/s. The wind speed for Barisal was above 2.40 m/s during the period, while the speed for Sylhet was consistently below 1.50 m/s. The wind speeds of Dhaka, Rangpur, and Rajshahi varied mostly between 1.81 and 2.10 m/s. These regions correspond to regions with higher atmospheric CO₂ concentrations, as shown in Fig. 8(c).

3.7. Correlation analysis

Correlation analysis was performed in this study to have a better understanding of the scenario and to check if the relevant factors contribute to the condition. Both the Pearson correlation coefficient (r_p) and the Spearman's Rank Correlation Coefficient (r_s) were determined. The outcome of the correlation analysis between MXCO₂ and the relevant factors is shown in Fig. (9).

The correlation analysis results in Fig. (9) show that MXCO₂ has a strong positive correlation with anthropogenic CO₂ emission, as shown in Fig. 9(a). The correlation between Tropospheric NO₂ and MXCO₂ is also strongly positive, as observed in Fig. 9(b). Furthermore, the population density also exhibited a high positive correlation with MXCO₂ in Fig. 9(c). Fig. 9(d) and (e) illustrate the correlation of MXCO₂ with NDBI and NDVI, respectively, where a moderate correlation was observed for both factors. The correlation with NDBI was positive, while the correlation with NDVI was negative. The correlation of MXCO₂ with the three meteorological variables, namely rainfall, temperature, and wind speed, are in Fig. 9(f), Fig. 9(g), and Fig. 9(h), respectively. The correlation of

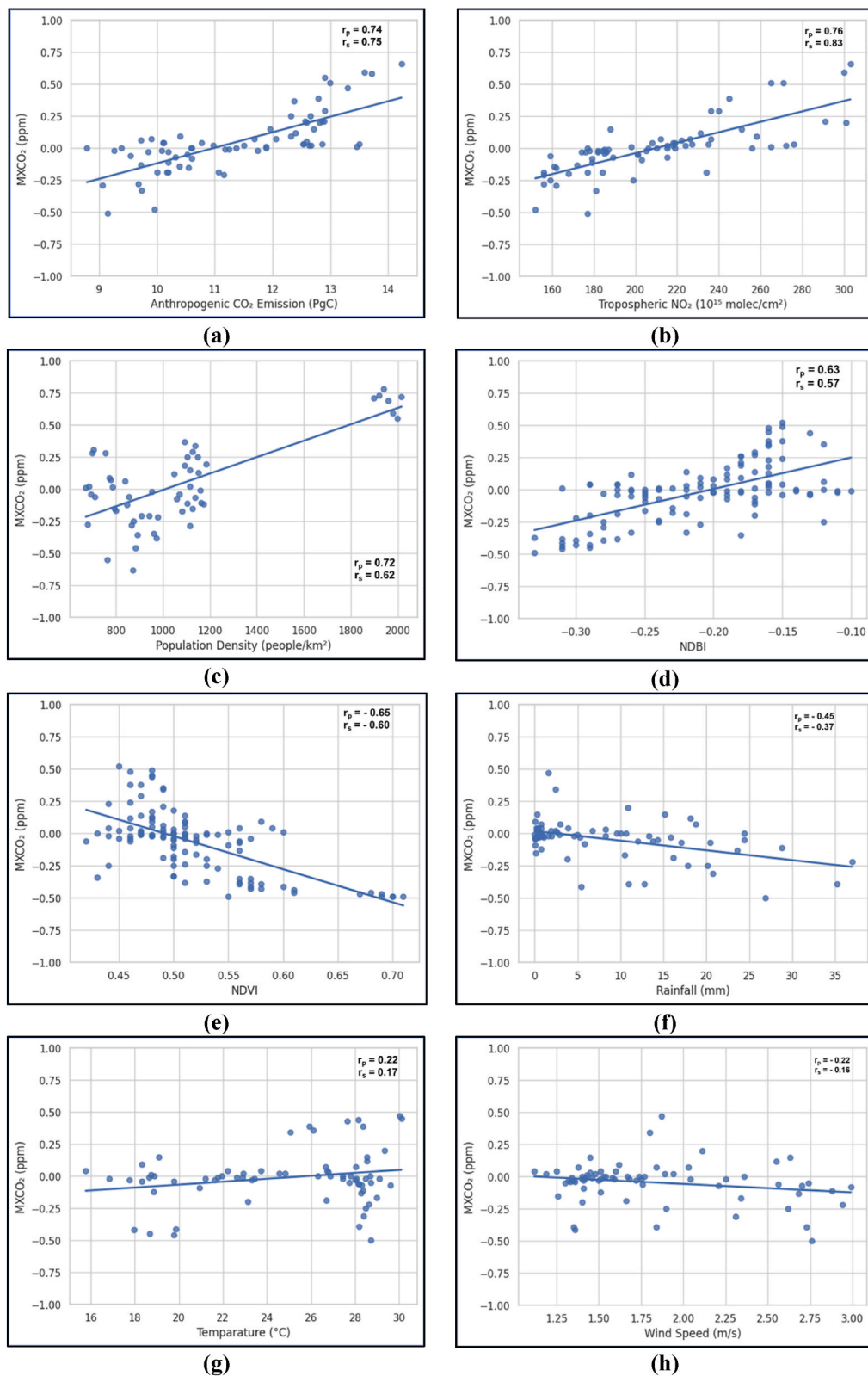


Fig. 9. Scatterplots of correlation analysis: (a) anthropogenic CO₂ emission and MXCO₂, (b) tropospheric NO₂ and MXCO₂, (c) population density and MXCO₂, (d) NDBI and MXCO₂, (e) NDVI and MXCO₂, (f) rainfall and MXCO₂, (g) temperature and MXCO₂, (h) wind speed and MXCO₂.

MXCO₂ with rainfall was moderately negative. A weak positive correlation with temperature and a weak negative correlation with wind speed was observed for MXCO₂.

3.8. Climate-vulnerable regions

Climate-vulnerable regions were identified based on the rate of increase in atmospheric concentration and anthropogenic emission of CO₂, a key driver of climate change. Table 1 presents the rates of increment for the two factors over the study period. Among the eight divisions analyzed, the Dhaka Division exhibited the highest growth rates in both atmospheric CO₂ concentrations (4.16 %) and anthropogenic emissions (28.96 %), signifying it as the most vulnerable region. In contrast, the Barishal division exhibited the lowest growth rates in CO₂ concentrations (3.13 %) and anthropogenic emissions (28.20 %), suggesting the lowest vulnerability to climate change impacts. Divisions exhibiting higher urbanization rates and lower vegetation cover demonstrated greater vulnerability. Rangpur and Rajshahi were ranked second and third, respectively, in terms of overall vulnerability. Rangpur showed a higher rate of increase in terms of atmospheric CO₂ concentration than Rajshahi. Although the increment rate of CO₂ concentration of Sylhet was higher than that of Mymensingh and Khulna, the increment rate of anthropogenic emission was much lower for Sylhet.

4. Discussion

Atmospheric CO₂ concentrations have exhibited a long-term increasing trend across all the divisions of Bangladesh over the years with seasonal fluctuations. The spatio-temporal distribution of XCO₂ showed the highest concentration for Dhaka and the lowest concentration for Barishal. The distribution pattern of anthropogenic CO₂ emission is almost similar, which complements the results. Dhaka division, encompassing the densely populated capital city and industrial centers such as Gazipur and Narayanganj, experienced elevated CO₂ concentrations. Several factors contribute to this increase, including rapid and often unplanned industrialization and urbanization, the proliferation of brick kilns, heavy traffic congestion, and ongoing climate change [65]. CO₂ concentrations showed a seasonal trend, peaking during the summer months, particularly in April. Conversely, concentrations decreased significantly during the rainy season, reaching the lowest value in August throughout the study period. In summer, higher temperatures and radiation promote the assimilation of atmospheric CO₂ during the day but increase respiration at night [66]. The lower wind speed during the pre-monsoon season is also a potential contributor to increased atmospheric CO₂ [67]. Among several reasons, the reduction in CO₂ concentration primarily occurs due to the strong photosynthesis process and weak respiration rate of plants that is enhanced during the monsoon. Conversely, after the monsoon, diminished photosynthesis and increased respiration result in a rise in CO₂ concentrations [3]. Bangladesh experiences high temperatures from April to June, coinciding with the observed peak in CO₂ concentrations. Furthermore, the country receives approximately 80 % of its annual rainfall between June and September, directly aligning with the observed decrease in CO₂ during the monsoon season [68]. These climatic patterns provide a strong contextual basis for the seasonal CO₂ trends observed. The MXCO₂ distribution of Dhaka indicated an increasing concentration and emission scenario. Rajshahi showed greater CO₂ absorption compared to other areas. This is likely because, despite its size, the anthropogenic CO₂ emissions in the area were relatively low. The drastic change in MXCO₂ in Khulna in 2021 can be attributed to urban expansion in recent times [69].

The spatio-temporal distribution of atmospheric CO₂ concentrations showed relative similarities with other factors. Anthropogenic CO₂ emissions also showed an increasing trend during the period. In 2020, the amount of emissions was reduced for the whole country. Although the reduction was minimal, this incident goes against the overall increasing trend as the emissions increased again in 2021 at a higher amount than that of 2019. It can be associated with the COVID-19 pandemic lockdown, when industries were shut down and fossil fuel combustion was reduced significantly for certain periods [70]. The regions with higher population density exhibited higher CO₂ concentrations and emissions, a pattern frequently observed in developing nations [71]. The NDVI trend declined in regions with higher CO₂ concentrations, indicating the role of vegetation in the CO₂ sink. Conversely, the regions with higher NDBI trends were mostly associated with areas of higher CO₂ concentrations and anthropogenic CO₂ emissions, which indicates the effects of unplanned urbanization. These findings are consistent with previous research [72], which demonstrates that land-use changes and urbanization contribute to increased CO₂ concentrations, potentially overwhelming the natural carbon sequestration capacity of the region. While no distinct trends in meteorological variables were observed during this short timeframe, annual mean rainfall was comparatively higher in regions with lower atmospheric CO₂ concentrations.

Table 1
Climate-vulnerable regions ranking.

Division	Rate of Increment (%)		Vulnerable Regions Rank
	XCO ₂ Concentration	Anthropogenic CO ₂ Emission	
Dhaka	4.16	28.96	1
Rangpur	4.13	28.76	2
Rajshahi	3.82	28.78	3
Mymensingh	3.65	28.73	4
Sylhet	3.84	28.48	5
Khulna	3.35	28.70	6
Chattogram	3.26	28.46	7
Barishal	3.13	28.20	8

The correlation analysis between MXCO_2 and the other factors provided insights into its spatial distribution patterns. Anthropogenic CO_2 emissions and mean tropospheric NO_2 exhibited a strong positive correlation with MXCO_2 . These results indicate that anthropogenic emissions from fossil fuel combustion significantly contribute to the rising atmospheric CO_2 concentrations. Hakkarainen et al. [54] found similar results for the Southeast Asian regions. The strong positive correlation with population density further supports the contribution of anthropogenic emissions. The negative correlation with NDVI and positive correlation with NDBI help to explain the similarity of the distribution pattern described earlier and highlight the potential negative impacts of unplanned urbanization within the country. The meteorological variables had a weaker correlation with MXCO_2 individually. This can be due to the complex relationship between CO_2 concentrations and meteorological variables and the influence of multiple factors directly or indirectly, which contribute simultaneously to the situation [73]. These findings are specific to the study area and period during which no discernible trends were observed. Finally, based on CO_2 concentration and emission rates during the study period, the vulnerability ranking of the divisions declared Dhaka the most vulnerable region and Barishal the least vulnerable region. Dhaka is a highly populated and urbanized area that serves as the main industrial region of the country [74], where CO_2 concentrations and emissions are higher than in other areas. At the same time, Barishal is the least industrialized and moderately urbanized region [75]. The distribution patterns of the other factors further support the findings for better understanding. Highly vulnerable regions were characterized by metropolitan areas with large-scale economic activities. Thus, the population growth and land-use change driven by industrialization and rapid urbanization necessitate special attention to tackle the increasing CO_2 concentration.

5. Conclusion

This study provides a comprehensive analysis of the spatio-temporal distribution of atmospheric CO_2 concentrations over Bangladesh. The findings indicate that Dhaka was the region with the highest CO_2 concentrations and emissions. The CO_2 concentrations for the whole country were observed to be higher during the summer and lower during the monsoon season. Anthropogenic CO_2 emission was identified as the primary driver of observed CO_2 concentration. The higher correlation with tropospheric NO_2 , population density, decreasing NDVI trend, and increasing NDBI trend supports this finding. Finally, Dhaka emerged as the most vulnerable region, while Barishal exhibited the least vulnerability. To address the higher CO_2 concentrations and anthropogenic emissions in Dhaka, the decentralization of industrial activities could be considered. Implementing sustainable land-use practices, renewable energy policies, afforestation, and green infrastructure can help tackle haphazard urbanization, declining vegetation, and increasing CO_2 concentrations and emissions. The results of this study show the effectiveness of the remote sensing approach in observing atmospheric CO_2 . Urban planners, environmental scientists, and policymakers can consider the findings of the study for adopting effective mitigative measures and sustainable urban planning. The study period is limited due to the availability and consistent spatio-temporal coverage of datasets. Future research could benefit from extending this period using larger datasets to obtain more robust and long-term insights. The datasets from recently launched GOSAT-2 and OCO-3 satellites can be utilized to mitigate spatio-temporal data gaps.

CRedit authorship contribution statement

Bhaskar Ghosh: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Md. Zunaid Farouque:** Writing – review & editing, Visualization, Validation, Software, Resources, Formal analysis, Data curation. **Tanmoy Chakraborty:** Writing – review & editing, Validation, Supervision.

Data availability statement

Data included in the article/supplementary material is referenced in the “Methodology” section of the article.

Ethical approval

Review and/or approval by an ethics committee was not needed for this study because this study is based on secondary data.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.heliyon.2025.e44434>.

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